Practice Problems
Calculus Physics II

**Name:** Key **Date:** _____

## Chapter 56 – Polarization of EM Waves

1) For a beam of light, the direction of polarization is defined as

A) the beam's direction of travel.
B) the direction of the electric field's vibration.
C) the direction of the magnetic field's vibration.
D) the direction that is perpendicular to both the electric and magnetic field vectors.

2) If unpolarized light of intensity $I_o$ passes through an ideal polarizer, what is the intensity of the emerging light?

A) $I_0$
B) $I_0/2$
C) $I_0/4$
D) $I_0/\sqrt{2}$
E) $I_0/16$

$I = \frac{1}{2} I_0$ For unpolarized light

3) For unpolarized light incident on a single polarizer that is orientation with its transmission axis rotated 30° from the vertical, what is the orientation of the polarized wave after the polarizer?

A) Vertical
B) Horizontal
C) 60° to the vertical
D) 45° to the vertical
E) 30° to the vertical

4) If the transmission axis of two polarizers are at 90° to each other, how much light is transmitted through both polarizers if there is unpolarized light incident on the polarizers?

A) All light is transmitted.
B) No light is transmitted.
C) Half the light is transmitted.
D) 1/3 of the light is transmitted.

5) If the transmission axis of two polarizers are at 0° to each other, how much light is transmitted through both polarizers if there is polarized light incident on the polarizers is oriented at 90° to the transmission axis of the first polarizer?

A) All light is transmitted.
B) No light is transmitted.
C) Half the light is transmitted.
D) 1/3 of the light is transmitted.

6) If polarized light at 20° to the vertical is incident on a polarizer that has it's transmission oriented at 50° to the vertical in the same direction as the light, what angle would be used to calculate the transmitted intensity of the EM wave?

$\theta = 50° - 20° = 30°$

A) 50°
B) 30°
C) 20°
D) 80°
E) 10°

7) Unpolarized light with an intensity of 400 W/m$^2$ is incident on a polarizer with its transmission axis vertical, what is the intensity of the transmitted light and what is the polarization orientation of the transmitted light?

$I = \frac{1}{2} I_0 = 200 \text{ W/m}^2$

8) Polarized light of intensity $I_o$ passes through an ideal polarizer. If the electric vector of the polarized light is horizontal what, in terms of the initial intensity $I_o$, is the intensity of the light that passes through a polarizer if that polarizer is tilted 25° from the horizontal?

$I_1 = I_0 \cos(25°)^2 = (0.82) I_0$

A) 0.81 $I_o$
B) 0.19 $I_o$
C) 0.22 $I_o$
D) 0.28 $I_o$

9) Un-polarized light with an intensity of 400 W/m$^2$ is incident on a polarizer with its transmission axis vertical, then it goes through a second polarizers with the transmission axis at 70° relative to the vertical, what is the intensity of the light exiting the second polarizer?

$I_1 = \frac{1}{2} I_0 = 200 \text{ W/m}^2$

$I_2 = I_1 \cos(70°) = 234 \text{ W/m}^2$

10) Vertically polarized light with an intensity of 1,000 W/m$^2$ is incident on a polarizer with its transmission axis at 20° to the vertical, then it goes through a second polarizers with the transmission axis at 60° relative to the vertical, what is the intensity of the light exiting the second polarizer?

$I_1 = I_0 \cos(20)^2$

$I_1 = 883 \text{ W/m}^2$

angle between polarized light + 2nd polarizer

$\theta_2 = 60° - 20° = 40°$

$I_2 = I_1 \cos(40°)^2$

$I_2 = 518 \text{ W/m}^2$

11) Vertically polarized light with an intensity of 600 W/m$^2$ is incident on a polarizer with its transmission axis at 30° to the vertical, then it goes through a second polarizers with the transmission axis at 90° relative to the vertical, what is the intensity of the light exiting the second polarizer?

$I_1 = I_0 \cos(30°)^2$

$I_1 = 450 \text{ W/m}^2$

$\theta_2 = 90° - 30°$

$\theta_2 = 60°$

$I_2 = I_1 \cos(60°)^2$

$I_2 = 112.5 \text{ W/m}^2$

12) A vertically polarized beam of light of intensity 100 W/m$^2$ passes through three polarizers. The first polarizer's transmission axis is making an angle of 20° with the vertical and the second polarizer's transmission axis is making an angle of 50° with vertical. Then the light beam is incident on the third polarizer that is rotated back to 20° What is the transmitted intensity of this beam of light through the three polarizers?

$I_1 = I_0 \cos(20°)^2$

$I_1 = 88.3\ W/m^2$

$\theta_2 = 50° - 20°$

$\theta_2 = 30°$

$I_2 = I_1 \cos(30°)^2$

$I_2 = 66.2\ W/m^2$

$\theta_3 = 50° - 20° = 30°$

$I_3 = I_2 \cos(30°)^2$

$I_3 = 49.7\ W/m^2$

13) A vertically polarized beam of light of intensity 100 W/m$^2$ passes through two ideal polarizers. The transmission axis of the first polarizer makes an angle of 20° with the vertical, and the transmission axis of the second one makes an angle of 110° with the vertical. What is the intensity of the light after it has passes through both polarizers?

$I_1 = I_0 \cos(20°)^2$

$I_1 = 88.3\ W/m^2$

$\theta_2 = 110° - 20°$

$\theta_2 = 90°$

$I_2 = I_1 \cos(90°)^2$

$I_2 = 0\ W/m^2$

14) Vertically polarized light with $I_1$ = 200 W/m$^2$ is incident upon a polarizing with its transmission angle Θ relative to the vertical, the transmitted light has a Intensity of $I_2$ = 120 W/m$^2$, what angle does the transmission axis make to the vertical Θ?

$I_2 = I_1 \cos^2(\theta)$

$\cos^2(\theta) = \frac{I_2}{I_1}$

$\cos(\theta) = \sqrt{\frac{I_2}{I_1}}$

$\theta = \cos^{-1}\left(\sqrt{\frac{I_2}{I_1}}\right)$

$\theta = 39.2°$

15) Brewster's angle is the angle in which the reflected light is

A) Totally internally reflected.
B) Completely polarized.
C) Bent at angle theta.
D) Refracted.

KEY

1) C
2) B
3) E
4) B
5) B
6) B
7) 200 $W/m^2$
8) A
9) 23.4 $W/m^2$
10) 518 $W/m^2$
11) 112.5 $W/m^2$
12) 49.7 $W/m^2$
13) 0 $W/m^2$
14) 39°
15) B